AI Use Disclosure: Dune Sentiment Analysis in R

Student: Nick | AI tool: Claude (Anthropic), Claude Opus 5.5 | Sessions: November 7–8, 2026

Summary

I used Claude as a coding assistant and tutor while doing a tidytext sentiment analysis of three Frank Herbert novels (*Dune*, *Dune Messiah*, *Children of Dune*) in R and Quarto. I mostly used it to explain R functions, debug my own code, interpret results, and draft custom sentiment lexicons. I wrote the analysis pipeline, ran all of the code in my own R session, and produced the plots and results myself. Claude could not run R in its environment, so none of the numbers or figures in my assignment were generated by the AI.

Log of interactions

Topic / my question	What the AI provided	Type of help
How `lead()` and `lag()` work in R	Explanation with examples; warned about base R `stats::lag()` masking and the need to sort rows first.	Explanation
My `mutate()` code to number chapters from `line_num` resets	Identified syntax errors in my `ifelse()`/`cumsum()` call and showed a corrected version using `cumsum()` on a logical condition.	Debugging
Running count that increases each time a value is TRUE	Showed the `cumsum(condition)` pattern, grouping, and the SQL window-function equivalent.	Explanation
My `cumsum()` returning all NAs	Explained NA propagation from the first-row `lag()`; suggested `coalesce()` or a `default` argument.	Debugging
Error from `str_detect("=")`	Explained the missing `string` argument and showed how to detect chapter-divider lines.	Debugging
Appending three datasets	Explained `bind_rows()` with an `.id` column to label each book.	Explanation
Breakdown of a Bing sentiment pipeline (join, count by 80-line chunk, pivot, net score)	Step-by-step explanation of what each line does, plus caveats about chunk size and context-free lexicons.	Explanation
A Dune-specific weighted sentiment lexicon	Drafted an R script with about 180 words weighted -5 to +5, including Dune terms (e.g. water, sietch, Harkonnen) and neutral overrides for words Bing misreads (desert, worm). Weights are the AI's judgement calls.	Generated code
A Dune-specific emotion lexicon for a radar chart	Drafted an R script sorting about 225 words into six buckets (fear, wrath, treachery, faith, endurance, devotion) plus radar-chart code with a `coord_radar()` helper.	Generated code
Whether my NRC proportion code was correct	Found a bug (dividing the text column instead of n) and suggested filtering out NRC's positive/negative categories.	Debugging
Why my three books' NRC radar profiles look nearly identical	Interpretation: large samples, shared authorial style, and scale compression. Also flagged a double-percent axis bug and suggested difference-from-average plots, a chi-square test, and within-book trajectories.	Interpretation
Inline R values in Quarto text, and a coverage-percentage sentence	Explained code chunks, chunk options, inline `` `r ...` `` code, `source()`, and `semi_join()` to count rated words without NRC duplicates.	Explanation

Topic / my question	What the AI provided	Type of help
Hiding messages and warnings in Quarto	Showed the global `execute:` YAML options.	Explanation

What was mine and what came from the AI

- **My work:** choosing the texts and research question, loading and tokenizing the books, writing the original pipeline code, running every analysis, producing the figures, and deciding what to include in the write-up.
- **AI-assisted:** fixes to specific lines of my code, explanations of R and Quarto functions, and suggestions for interpreting and improving my visualizations.
- **AI-generated:** the two custom Dune lexicon scripts (word lists, weights, and emotion buckets) and the radar-chart helper code. I treated these as starting drafts to check against my data.

Limitations and checks

- Claude could not execute R code during the sessions, so all code it suggested had to be run and checked in my own environment.
- The custom lexicon word lists and weights reflect the AI's interpretation of the books, not a validated lexicon. Any results using them should be read with that in mind.
- Interpretation of my results (e.g. why the NRC profiles look similar) was offered as explanation; conclusions in the assignment are my own.